

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

Assessment Tool Weightage: 40%

QUESTIONS: 05 Total Marks:50

Annexure A

Industry Problem Solving Task

<i>Criteria</i>	Understandi ng of Industry Problem (2)	Applicati on of Engineeri ng	Soluti on Desig n	Use of Tools (2)	Analysis (1)	<i>Total(10)</i>
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		Concepts (2.5)	(2.5)			
Weightage	20%	25%	25%	20%	10%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Code

of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action. **Signature**

of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem	Good understanding with minor gaps in requirements	Basic understanding; some requirements	Poor understanding; misinterpretations problem

		and constraint s	nt s	unclear	context
Applicati on of Engineeri ng Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropria te concepts with minor errors	Limited applicati on of concepts	Incorrect or no applicati on of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impracti cal solution
Use of Tools	15%	Effective use of modern tools, technologi es , and real data	Appropria te use of tools with correct implement at ion	Limited or inconsiste nt use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performan ce evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Profession ali sm	5%	Highly profession al, well-docu me nted, and clearly	Good document ati on and presentation	Basic document ati on with some gaps	Poor document ati on and presentat io n

		presented work			
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Annexure A

Q1 . Dataset: Supermarket Purchases

Customer Visits/Month Avg Spend (₹) Items Bought

C1 10 2000 15

C2 4 5000 8

C3 12 1800 18

C4 3 6000 6

C5 11 2100 16

Question:

Using the above dataset, propose a clustering approach to segment customers and explain how it supports targeted marketing. *(BL3 – 7 Marks)*

Q2. Dataset: Credit Card Transactions

Transaction Amount (₹) Location Time (hrs)

T1 200 Chennai 10

T2 5000 Delhi 2

T3 150 Chennai 11

T4 7000 Mumbai 1

T5 180 Chennai 12

Question:

Design a clustering-based solution using this dataset to detect unusual or fraudulent transactions. *(BL4 – 7 Marks)*

Q3. Dataset: Telecom Usage

User Call Duration (min) Data Usage (GB) Recharge (₹)

U1 300 5 200

U2 100 2 100

U3 350 6 220

U4 80 1 90

U5 320 5.5 210

Question:

Explain how clustering can be applied to identify churn-prone users and improve customer retention strategies. *(BL4 – 7 Marks)*

Q4 . Dataset: E-commerce Behavior

User Product Views Clicks Purchases

U1 50 20 5

U2 30 10 2

User Product Views Clicks Purchases

U3 60 25 6

U4 20 8 1

U5 55 22 5

Question:

Suggest a clustering technique to group users and explain how it improves recommendation systems. (BL3 – 7 Marks)

Q5. Dataset: Traffic Data

Location Vehicle Count Avg Speed (km/h) Delay (min)

L1 200 30 10

L2 500 15 25

L3 220 28 12

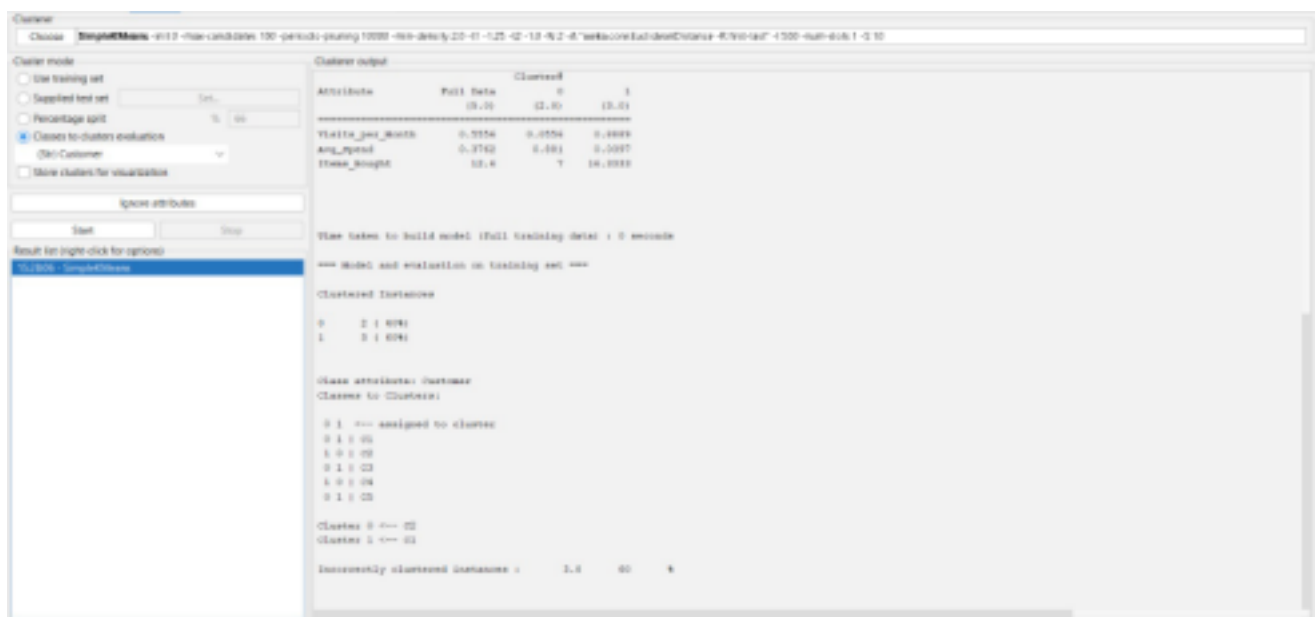
L4 600 10 30

L5 210 32 9

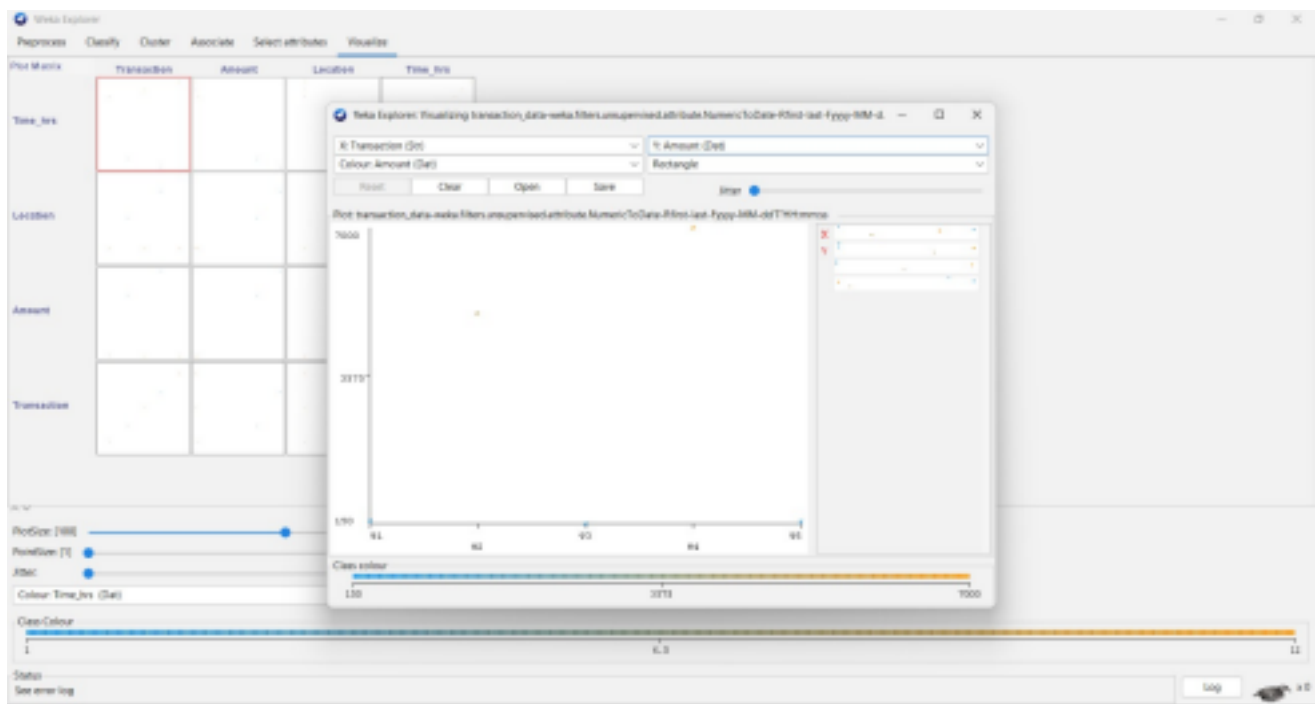
Question:

Design a clustering-based solution to identify traffic congestion zones and explain how it helps in traffic management. (BL4 – 7 Marks)

Q1 - Answer



Q2 - Answer



Q3 - Answer

```

Attribute          Full Data      Classes#
                  (5.9)      0      1
=====
Call_Duration_min    230      90  323.3333
Data_Usage_MB        3.9      1.5   5.5
Package              164      85  210

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0      2 ( 40%)
1      3 ( 60%)

Class attribute: User
Classes to Clusters:

0 1 <-- assigned to cluster
0 1 | U1
1 0 | U2
0 1 | U3
1 0 | U4
0 1 | U5

Cluster 0 <== U1
Cluster 1 <== U1

Incorrectly clustered instances :      3.0      60      %

```

Q4 - Answer

Choose **SimpleKMeans** with 8 max-candidates 100 periods pruning 10000 min-density 20 -k 1-25 -q 1-0-M2 -A 'wikiscore(EuclideanDistance 0.5-trimmed)' -i 500 -num-obj 1-5 50

Cluster mode

- ☐ Use training set
- ☐ Supplied test set
- ☐ Percentage split
- ☒ Classes to cluster evaluation
- ☐ (Str) Use
- ☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

10/10/20 SimpleKMeans

Cluster output

Final clusters centroids:

Attribute	Full Data	Cluster#	0	1
	(15, 0)	(12, 0)	(15, 0)	
=====				
Product_View	0.575	0.325	0.575	
Clicks	0.3294	0.0000	0.0431	
Purchases	0.0	0.5	0.0000	

Time taken to build model (Full training detail) : 0.00 seconds

=== Model and evaluation on training set ===

Clustered Instances

```

0  2  ( 40%)
1  3  ( 40%)

```

Class attributes: Over
Classes to clusters:

```

0 0 --- assigned to cluster
0 1 01
1 0 00
0 1 10
1 0 10
0 1 00

```

Cluster: 0 --- 00
Cluster: 1 --- 00

Incorrectly clustered instances : 3.0 60 4

Q5 - Answer

Choose **SimpleKMeans** with 8 max-candidates 100 periods pruning 10000 min-density 20 -k 1-25 -q 1-0-M2 -A 'wikiscore(EuclideanDistance 0.5-trimmed)' -i 500 -num-obj 1-5 50

Cluster mode

- ☐ Use training set
- ☐ Supplied test set
- ☐ Percentage split
- ☒ Classes to cluster evaluation
- ☐ (Str) Use
- ☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

10/10/20 SimpleKMeans

Cluster output

Final clusters centroids:

Attribute	Full Data	Cluster#	0	1
	(15, 0)	(12, 0)	(15, 0)	
=====				
Vehicle_Count	0.365	0.875	0.000	
Avg_Speed_Mph	0.1805	0.016	0.000	
Delay_mins	17.2	27.5	10.000	

Time taken to build model (Full training detail) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

```

0  2  ( 40%)
1  3  ( 40%)

```

Class attributes: location
Classes to clusters:

```

0 1 --- assigned to cluster
0 1 11
1 0 00
0 1 11
1 0 00
0 1 11

```

Cluster: 0 --- 11
Cluster: 1 --- 00

Incorrectly clustered instances : 3.0 60 4